Fast Simulation of Inextensible Hair and Fur

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Introduction

- Müller, Matthias & Kim, Tae & Chentanez, Nuttapong. (2012). Fast Simulation of Inextensible Hair and Fur
- Simulate hair or fur on humans or animals
- How it's done:
  - Updates position of strands
  - Uses follow the leader to reposition strands with constraint conflicts
  - Updates velocity of strands derived from their positions with corrections for uneven mass distribution

Figure 1: *Our method allows the simulation of every hair strand in real time. From left to right: 47k hairs simulated at 25 fps including rendering and hair-hair repulsion. Long hair composed of 1.9m particles at 8 fps. Curly hair using visualization post-processing.*

How It Is Done

- Built on a combination of prior research
  - Follow the Leader for constraint violations
  - Regular background grid for for repulsion and friction
  - Position Based Dynamics to store and update velocities
- Steps
  - Updates position of strands
  - Reposition strands with constraint conflicts
  - Updates velocity of strands derived from their positions
  - Update velocity further for hair/hair friction and repulsion

Problem

- Realistic simulations of hair/fur are computationally expensive
- Existing methods settle for inextensibility or increasing the amount of frames it takes to compute
- A solution is needed that has all these attributes
  - Thousands of hairs
  - Inextensibility
  - 1 iteration to complete
  - Stable behavior
  - Realistic collision with character skin

Real World Applications

- Computer games
- Movies and television
- Mixed reality
- 3D modeling art

Contribution of the Paper

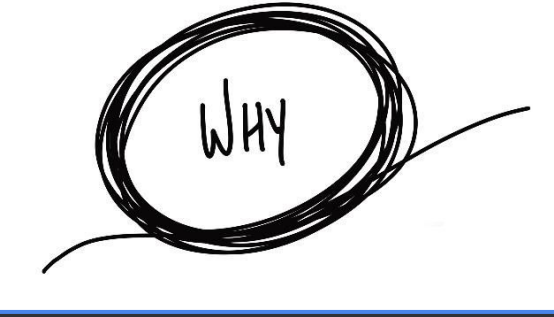
- Combining existing solutions to create one that has:
  - Inextensibility
  - Single iteration
  - Thousands of strands of hair
  - Stable behavior
  - Simulation of each strand of hair in real time
- Main contribution
  - Fixed issue of uneven mass distribution during dynamic FTL
  - $v = ((p_i - x_i) / \Delta t) + s_{damping} * (-d_{i+1} / \Delta t)$

How are the results evaluated?

Real-time, Visually plausible, and Physically Stable hair motion.

- Zero-stretch (inextensibility)
- Smooth and stable motion
- Realistic strand behavior (bending, shaping, no jittering)
- Natural interaction (wind/blower, collisions)
- Proper hair-hair behavior (repulsion vs clumping)

Fig. 3. Captures from our hair ball simulations, from top to bottom: 16k, no continuous-time; 16k, continuous-time; 128k, no continuous-time; 128k, continuous-time. Skipping continuous-time collision detection leads to a more clumped final state, as strands that get entangled due to missed collisions during the high-speed impact phases maintain proximity and cannot separate again at the later stages.



1. Performance Metrics

The FPS + particle/strand counts demonstrate how large and dense the simulations can get while still running interactively.

- 47k hairs at 25 FPS (handle game-level, high-density hair in real time)
- 1.9M particles at 8 FPS (scales very long hair while remaining interactive.)

The method meets the goal of real-time, visually plausible, stable hair motion even at large scales.

2. Solver Comparison

- DFTL: zero stretch + stable with one iteration
- PBD/Euler: stretching or instability unless greatly increasing iterations (which costs time)

These comparisons validate the method's ability to maintain zero-stretch, Smooth, stable motion, Realistic strand behavior

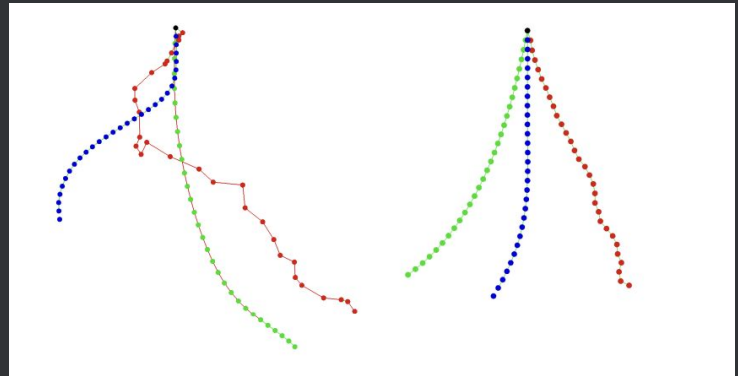


Figure 6: *Screen shots of an animation in which the top particle of a rope is dragged around using three simulation methods: dynamic FTL (blue), PBD (green) and symplectic Euler (red). Left: all three methods spend the same amount of time per frame. Right: Time spend is adjusted to yield similar results.*

Visual Quality Evaluation

They also evaluate results qualitatively, by examining:

- Shape preservation (hair retains intended form)
- Smooth motion (no jittering/over-stretching)
- Realistic bending (long hair curves naturally)
- Wind/blower response (fur being blown)
- Hair-hair repulsion (volume vs clumping)

These visuals demonstrate real-world plausibility that complements the numeric results.

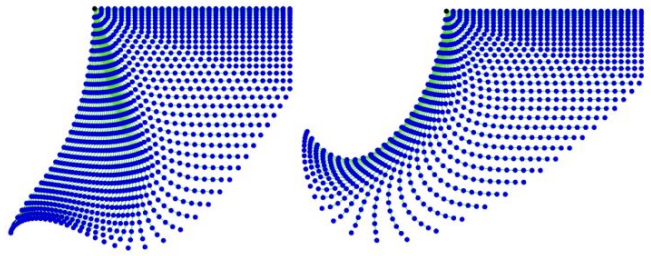


Figure 4: *Rope simulation with velocity correction. Left: Correction scale* $s_{damping} = 1$. *Right: Correction scale* $s_{damping} = 0.9$.

Strengths, and weaknesses

Strengths

- Zero-stretch hair simulation with only one iteration per frame
- Real-time performance for tens of thousands of strands
- Stable, smooth motion even under fast character movement
- Supports hair-hair repulsion, friction, and collisions
- Works for straight, long, curly hair and dense fur

Weaknesses

- Numerical damping due to geometric nature
- Less physically accurate than full rod/elastic simulations
- Simplified collision for character bodies
- Performance drops with very long, extremely dense hair
- Does not model torsion or complex bending stiffness

Future Work

- Improve physical accuracy by reducing numerical damping and adding richer strand physics (bending, torsion, twisting).
- Enhance interaction quality through more precise collision handling and more advanced hair-hair behavior (better volume preservation, less clumping).
- Increase scalability to support even denser or longer hair simulations while maintaining real-time performance.
- Live browser demo that exactly uses DFTL (Dynamic Follow-the-Leader) in WebG

Hair Mesh Rendering Demo